M500 In-Car Video System Installation Guide, International Version

AUGUST 2025

MN010315A01-C

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The Centralized Managed Support Operations (CMSO) is the primary contact for technical support included in your organization's service agreement with Motorola Solutions. To enable faster response time to customer issues, Motorola Solutions provides support from multiple countries around the world.

Service agreement customers should be sure to call the CMSO in all situations listed under Customer Responsibilities in their agreement, such as:

- To confirm troubleshooting results and analysis before taking action

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1. Enter motorolasolutions.com in your browser.
2. Ensure that your organization's country or region is displayed on the page. Clicking or tapping the name of the region provides a way to change it.
3. Select "Support" on the motorolasolutions.com page.

Documentation Portal

The Motorola Solutions (MSI) Documentation Portal is an online platform where you can find all the user documentation in one place, go to https://motr.la/docs.

You can provide feedback, questions, or comments for any publication or article by selecting the feedback icon. See Providing Feedback.

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Discover the many learning opportunities within the MSI Learning Center to explore your curiosities, shape your career and maximize your impact, go to https://learningcenter.motorolasolutions.com/.

Document History

Version	Description	Date
MN010315A01-A	Initial release of international version	September 2023
MN010315A01-B	RFID reader installation (RFID Installation on page 45) and updated wiring diagrams (Installation Overview UK on page 22) (Installation Overview EU on page 23)	February 2025
MN010315A01-C	The following sections were updated: ● Installation Overview UK on page 22 ● Installation Overview EU on page 23	August 2025

Supplier's Declaration of Conformity

Declaration of Conformity

Per FCC CFR 47 Part 2 Section 2.1077(a)

Responsible Party

Name: Motorola Solutions, Inc.

Model Name: M500

Address: 2000 Progress Pkwy, Schaumburg IL, 60196

Phone Number: 1-800-927-2744

Hereby declares that the product M500 conforms to the following regulations:

FCC Part 15, subpart B, section 15.107(a), 15.107(d), and section 15.09(a)

Class B Digital Device

As a personal computer peripheral, this device complies with Part 15 of the FCC Rules. Operation is subject to the following two conditions:

- This device may not cause harmful interference, and
- This device must accept any interference received, including interference that may cause undesired operation

NOTE: This equipment has been tested and found to comply with the limits for a Class B digital device, pursuant to part 15 of the FCC Rules. These limits are designed to provide reasonable protection against harmful interference in a residential installation. This equipment generates, uses and can radiate radio frequency energy and, if not installed and used in accordance with the instructions, may cause harmful interference to radio communications. However, there is no guarantee that interference will not occur in a particular installation.

If this equipment does cause harmful interference to radio or television reception, which can be determined by turning the equipment off and on, the user is encouraged to try to correct the interference by one of the following measures:

- Reorient or relocate the receiving antenna
- Increase the separation between the equipment and receiver
- Connect the equipment into an outlet on a circuit different from that to which the receiver is connected
- Consult the dealer or an experienced radio/TV technician for help

Intended Use

M500 is intended as a mobile device. To satisfy RF exposure requirements, the antenna must operate with a separation distance of at least 20 cm from all persons. Reference FCC KDB 784748, Section A.8.

Notice to Users (FCC and Industry Canada)

This device complies with Part 15 of the FCC rules and Industry Canada license-exempt RSS's per the following conditions:

- This device may not cause harmful interference.
- This device must accept any interference received, including interference that may cause undesired operation.
- Changes or modifications made to this device, not expressly approved by Motorola Solutions, could void the authority of the user to operate this equipment.

Simplified EU Declaration of Conformity

Motorola Solutions hereby declares that the radio equipment (M500) is in compliance with Directive 2014/53/EU. The full text of the EU Declaration of Conformity is available at the following link: http://www.motorolasolutions.com/emea-compliance via your local Motorola Solutions helpdesk, your dealer from where you purchased this radio or send an email to manufacturerdeclaration.eu@motorolasolutions.com.

This is a license exempt service, but restrictions of use may apply in some Non-EU countries.

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About This Document

Welcome to the Motorola Solutions *M500 In-Car Video System Installation Guide, International Version*. This guide is designed to provide basic information and instructions for installing the Motorola Solutions M500 in-car video system. The *M500 In-Car Video System Installation Guide, International Version* covers the following topics:

Preparing for a vehicle installation, including:

- Pre-installation checklist
- Recommended tools
- Installation best practices

Installing Motorola Solutions equipment in a vehicle, including:

- Recommended installation workflows
- M500 DVR and Display
- M5F, M5P, M5R, and M5E cameras
- Wireless radio
- Wireless Radio PoE Adapter
- LTE appliance
- Mobile data computer (MDC)
- Microphones
- Related components and cables
- Testing the installation

This document also includes an appendix about the M500 bracket information. The bracket information may be helpful for your installation.

The images in this document are representative of what you could experience while installing. They are meant to serve as a guide.

This guide contains general recommended instructions for installing a Motorola Solutions in-car video system in a vehicle. This document is not an exclusive or comprehensive blueprint for any particular vehicle installation. If you have a question about installing the system in a particular vehicle, contact Motorola Solutions Customer Service.

IMPORTANT: This document assumes knowledge of standard 12-volt vehicle installation best practices. It is meant to guide a technician through the specifics of installing Motorola Solutions equipment.

Related Documents and Information

For further information related to installing Motorola Solutions equipment in a vehicle that is not covered by the *M500 In-Car Video System Installation Guide, International Version*, see the following documents:

Document	Description
M500 Installation Overview and Quick Start Guide (UK)	This document combines the M500 wiring diagram and the quick start guide into one document. The wiring diagram shows you how and where the com-

Document	Description
	ponents of the M500 in-car video system connect to the backplane of the M500 DVR. The quick start guide provides basic information to get you started using the M500 DVR. This document is valid for the UK.
M500 Installation Overview and Quick Start Guide (EU)	This document combines the M500 wiring diagram and the quick start guide into one document. The wiring diagram shows you how and where the components of the M500 in-car video system connect to the backplane of the M500 DVR. The quick start guide provides basic information to get you started using the M500 DVR. This document is valid for the EU.
M500 In-Car Video System User Guide, International Version	This guide is designed to walk you through the basics of using your M500 to collect video and audio evidence. It provides instructions for using the M500 system during a shift and includes information about the basic components and operation of the M500 in-car system. This document is valid for international customers.
V300 Body-Worn Camera User Guide	This guide is designed to walk you through the basics of using your V300 body-worn camera to collect evidence. It provides instructions for using the camera during a shift and includes information about the basic components and operation of the camera.
V700 Body-Worn Camera User Guide	This guide is designed to walk you through the basics of using your V700 body-worn camera to collect evidence. It provides instructions for using the camera during a shift and includes information about the basic components and operation of the camera.

Chapter 1

Preparing for the Installation

This section includes information to help you prepare for the in-vehicle installation of the Motorola Solutions M500 in-car video system and related components. The section includes:

- **Pre-installation checklist**: Complete the items on this checklist before starting the installation (Before the Installation on page 15)
- **Recommended tools list**: Shows a list of tools, including some specialized items, you should have available before beginning the installation (Recommended Tools on page 15)
- **Installation required practices**: Lists a number of best practices that Motorola Solutions requires you to follow as you perform the vehicle installation (Installation Required Practices on page 16)

1.1

Before the Installation

Use the following list to help you prepare for a successful M500 system installation in a vehicle:

- Prepare to document the installation for a first-time vehicle (year and/or model), as applicable
- Gather all necessary tools for the installation (Recommended Tools on page 15)
- Remove any old video equipment, as applicable
- Verify that you have received all the components for the M500 system that you are installing

  NOTE: If you have any missing or damaged parts, contact Motorola Solutions Customer Service.

  - Make sure the mounting brackets are the correct type for the specific vehicle

  NOTE: When installing brackets, follow the instructions included with the bracket. If you need a copy of the bracket instructions, contact Motorola Solutions Customer Service.
- Determine the installation locations (installation plan) for all components (including brackets and cables)
  - Determine the wire connection points, for example, vehicle battery location, emergency light input, brake input, auxiliary inputs
- Roughly lay out the main components and cables in their installation locations to test positioning

  NOTE: If a cable is too short, contact Motorola Solutions Customer Service for a longer cable.
- Read through the Installation Required Practices (Installation Required Practices on page 16) and add them to your installation plan

  NOTE: Use the M500 installation overview wiring diagrams (Installation Overview UK on page 22 or Installation Overview EU on page 23) as a reference for the installation.

1.2

Recommended Tools

The following tools are recommended for installing the Motorola Solutions M500 in-car video system and related components in a vehicle:

- Drill, bits, and/or hole saws including 3/4-inch bit for antenna mounting
- 10-30 feet of 16-20 gauge primary wire for extending input cable connections, if necessary

- Wire strippers
- Wire crimpers
- Adjustable wrench that can be adjusted to 1-1/8th inch
- Various wrenches and sockets, including 24mm or 15/16-inch open-ended wrench
- Pliers
- Utility knife
- Torx® screwdrivers or bits, sizes T20, T15, T10, and T6
- SAE and metric Allen Wrench sets, sizes SAE-.050~3/8 | MM 0.7~10
- Electrical tape and/or heat-shrink tubing
- Zip ties
- Hand-held butane torch or lighter (optional)
- Loctite® 312 adhesive and primer or Equalizer® brand KlingOn™ Rearview Mirror Adhesive (KMK630)
- 3M™ VHB™ pre-cut tape or tesa® wire loom harness tape
- 3/16-inch x 11-foot fiberglass wire running kit
- Trim and molding tool set

1.3 Installation Required Practices

The following practices are required by Motorola Solutions to help you have a more successful installation experience.

1.3.1 Cables

The installation and proper handling of required cables ensure a successful installation of the Motorola Solutions M500 in-car video system.

1.3.1.1 HD-BNC Cables

The HD-BNC (high-definition BNC) cable connector uses a quick connect/disconnect locking mechanism.

Follow the following required practices when working with the HD-BNC cables:

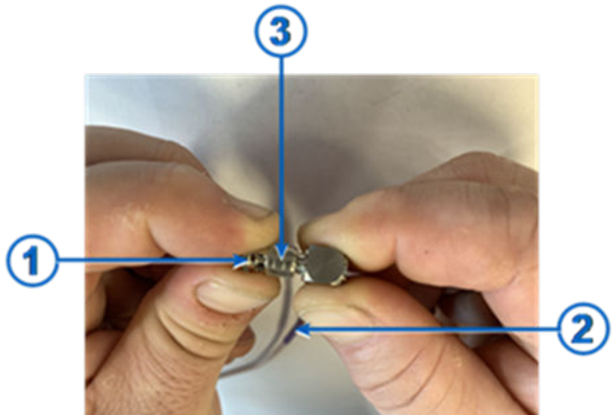
- Keep debris out of the connector when installing.
- Be careful of any sharp cable bends, kinks, or crimps. Use no more than a soft S-bend.

IMPORTANT: The HD-BNC cables included with the Motorola Solutions installation kits are customized to be as rugged as possible. However, HD-BNC cables are very susceptible to damage and do not work if kinked or pinched.

1.3.1.1.1
Connecting HD-BNC Cables

The HD-BNC connector uses a mini cylinder to lock the connector into the port.

Figure 1: HD-BNC Connector and a Cable



Item	Description
1	HD-BNC connector
2	HD-BNC cable
3	HD-BNC cylinder

Procedure:

1. Insert the HD-BNC connector into the jack port on the DVR.
2. While holding the HD-BNC cylinder, push towards the DVR and rotate clockwise until the cylinder locks in place.
3. Release the cylinder.
4. Verify that the connection is secure.

1.3.1.2
Pulling Cables

Pulling cables in the vehicle can be difficult and can damage the cables. Order and install new cables if you move a system from one vehicle to another.

NOTE: If any of the packaged cables have been damaged, contact Motorola Solutions Customer Service for a replacement.

1.3.2
Cameras

If you plan to mount the front camera on a puck using adhesive material, install the puck on the windshield first to allow the glue to dry.

TIP: To speed up the installation process, you can use a hand-held butane torch or lighter to heat the puck. For more information, see Installing the M5F Front Camera on page 25.

1.3.3
Connections

- Always butt-splice and crimp wire connections.
- Use 3M™ VHB™ pre-cut tape to cover any soldered electrical connections.

IMPORTANT: Do not use 3M™ ScotchLok™ or similar types of connectors.

1.3.4
Power

Directly connect power and ground wires to the appropriate battery posts. The battery is the cleanest source of power in the vehicle.

1.3.5
Wireless radio

Always mount the wireless radio in a position where you can easily see its LEDs. Access to the LEDs is needed for diagnostic purposes.

IMPORTANT: Do not mount the wireless radio in a location that is difficult to access, for example, in the console.

Chapter 2

M500 System Installation

This section includes information that will help you to install the Motorola Solutions M500 in-car video system and related components.

This section includes a recommended installation workflow (Installation Workflow on page 19).

This chapter also includes individual sections for installing the following components:

- M5F (Front), M5P, M5R, and M5E cameras (Camera Installation on page 24)
- M500 DVR (M500 DVR Installation on page 32) and its display (Installing the M500 DVR Display on page 32)
- Wireless radio and its antenna (Wireless Radio Installation on page 35)
- Wireless and cabin microphones (Microphone Installation on page 41)
- GPS antenna (Installing the GPS Antenna on page 42)
- LTE appliance (LTE Router Installation on page 43)
- MDC (MDC Installation on page 44)
- RFID reader (RFID Installation on page 45)
- Connecting the system to external inputs (Cable Connections for External Inputs on page 45)
- Connecting power to the M500 (Connecting the System Power Cable to the Vehicle Battery on page 47)
- Selecting the SPS Deployment option (Deployment Mode on page 49)

The final section provides instructions for the testing the M500 vehicle installation (Testing the System Installation on page 50).

2.1 Installation Workflow

The following steps make up a recommended workflow for installing the M500 in-car video system:

WARNING: Only connect cables when this workflow instructs you to. Connecting cables out of sequence can damage components.

Procedure:

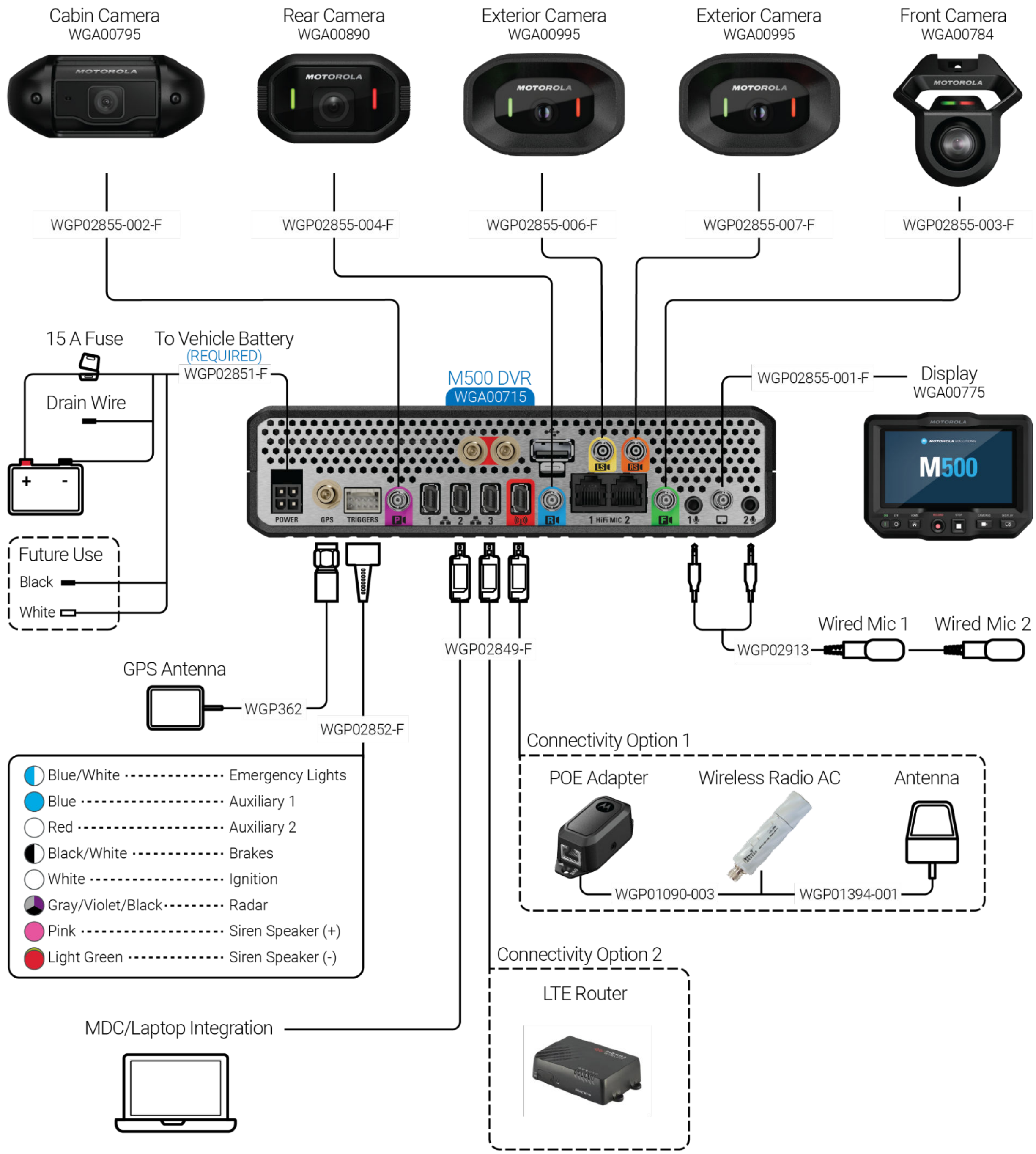
1. Complete all items in the Before Installing the M500 System in a Vehicle list (Before the Installation on page 15).
2. Remove vehicle panels as needed.
3. Remove cables from their packaging and roughly lay them throughout the vehicle to verify routing and length.
4. Install the **M5F front camera**:
    a. Mount the M5F front camera (Installing the M5F Front Camera on page 25).
    b. Connect the front camera high-definition BNC (HD-BNC) cable to the front camera then run it to the location where you will install the DVR, but DO NOT connect the cable to the DVR.
5. Install the **M500 DVR Display**:
    a. Mount the M500 DVR Display (Installing the M500 DVR Display on page 32).

   b. Run the display HD-BNC cable to the M500 DVR location, but DO NOT connect it to the DVR.

6. Install the **M500 DVR**:
   a. Install the M500 DVR bracket (M500 DVR Installation on page 32).
   b. Position and loosely install the DVR, but DO NOT connect cables to it.
7. If you have a wireless radio kit, position and install the **wireless radio** (but DO NOT connect cables) (Wireless Radio Installation on page 35).
8. If you have a wireless radio kit, install the **wireless radio antenna** and the radio-to-M500 Ethernet cable:
   a. Mount (drilled, magnetic, or trunk mount) the wireless radio antenna cable then run it to the wireless radio (Mounting the Wireless Radio Antenna on page 36).
   b. Connect the antenna cable to the wireless radio.
   c. Run the wireless radio Ethernet cable from the wireless radio to the red switch port on the back of the DVR (but DO NOT connect the cable).
9. (Optional): If you have a wireless radio kit, position and install the **wireless radio PoE adapter** and its related components (Installing the Wireless Radio PoE Adapter on page 39):
   a. Connect Ethernet cable 1 to the adapter port with the LEDs and connect the other end of the cable to the wireless radio.
   b. Connect Ethernet cable 2 to the adapter port marked **DVR** and run the other end of the Ethernet cable to the DVR location (but DO NOT connect the cable).
10. If you have an LTE router kit, position and install the **router** and its related components (LTE Router Installation on page 43):
    a. Connect an Ethernet cable from the appropriate port on the router.
    b. Run the other end of the Ethernet cable to the DVR location (but DO NOT connect the cable).
11. If you have an RFID kit, connect the RFID reader USB to the USB port on the DVR. (RFID Installation on page 45).
12. If you have an MDC kit, connect an Ethernet cable from the appropriate port on the MDC, then run the other end of the Ethernet cable to the DVR location (but DO NOT connect the cable).
13. Install the **stand-alone cabin microphones** and run their cables to the DVR location (but DO NOT connect the cables) (Installing the Stand-Alone Cabin Microphones on page 41).
14. Install the **GPS antenna** and run its cable to the DVR location (but DO NOT connect the cable) (Installing the GPS Antenna on page 42).
15. Install the **M5P** camera and run its cable to the DVR location (but DO NOT connect the cable) (Installing the M5P Camera on page 27).
16. If you have an **M5R** camera, install it and run its cable to the DVR location (but DO NOT connect the cable) (Installing the M5R Camera on page 29).
17. If you have one or more **M5E** cameras, install the cameras and run their cables to the DVR location (but DO NOT connect the cable) (Installing the M5E Camera on page 30).
18. Position and connect the **external inputs cable** (Cable Connections for External Inputs on page 45):
    a. Run the DVR connector end of the external inputs cable to the DVR location, but DO NOT connect the cable to the DVR.
    b. Run the sense wire ends of the external inputs cable to the best location for the vehicle, according to your installation plan.
    c. Connect each sense wire to the corresponding input wire, as needed (Cable Connections for External Inputs on page 45).

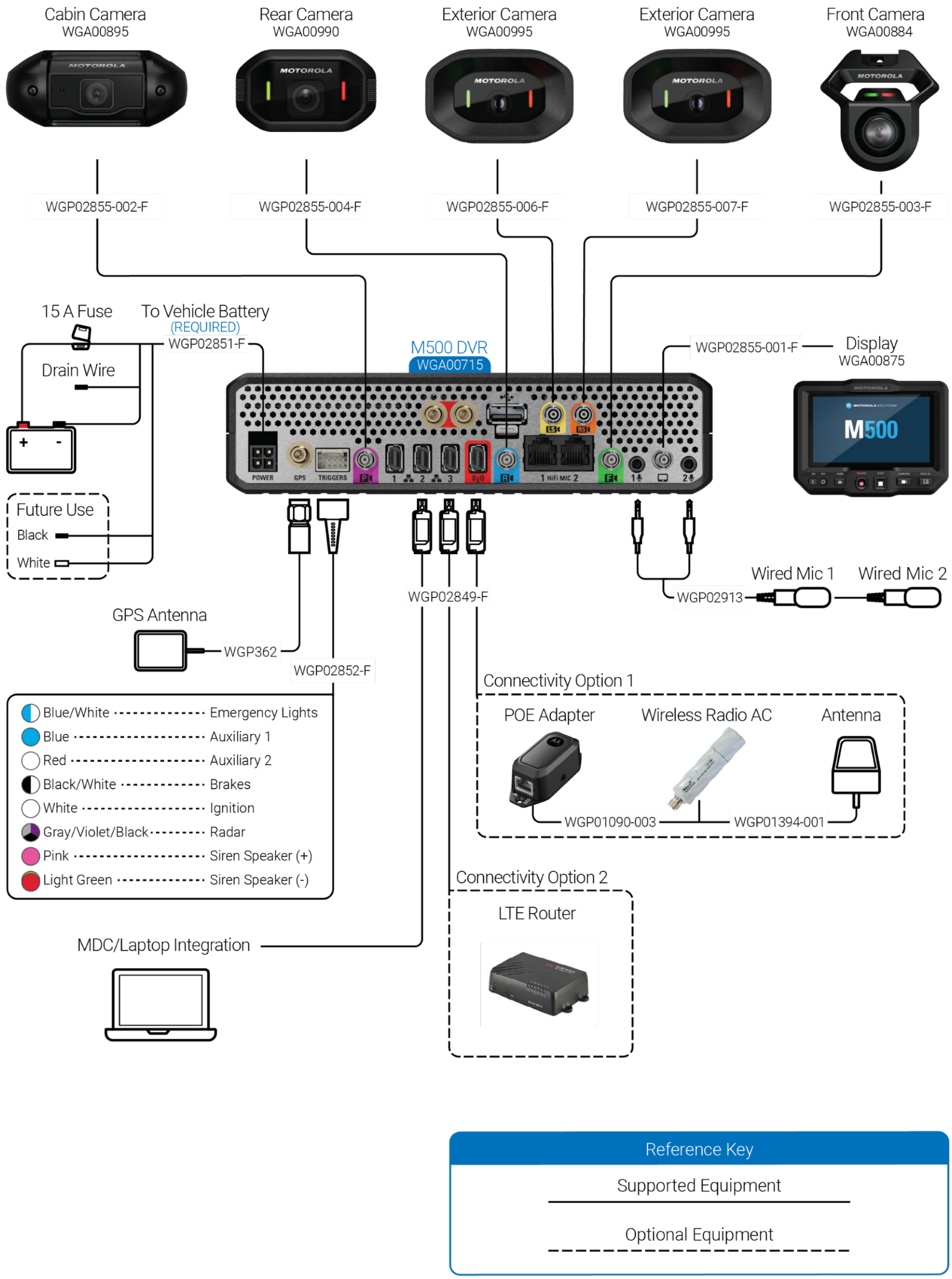
19. Install the **system power cable** (but DO NOT insert the 15 amp fuse in the fuse holder) (Connecting the System Power Cable to the Vehicle Battery on page 47):
    a. Run the DVR connector end of the power cable to the DVR location, but DO NOT connect the cable to the DVR.
    b. Run the power wire end of the power cable to the vehicle power source location and connect the cable to the power source.
20. Connect the following cables to the **DVR**, as applicable:
    a. Wireless radio Ethernet cable to the red switch port on the DVR or the PoE adapter Ethernet cable to one of the switch ports on the DVR
    b. LTE router Ethernet cable to one of the switch ports on the DVR
    c. MDC Ethernet cable to one of the switch ports on the DVR
    d. Cabin microphone cables to the appropriate connector ports
    e. Display HD-BNC cable to the appropriate connector port
    f. M5F, M5P, M5R, and M5E HD-BNC cables to their appropriate connector ports
    g. GPS cable to the **GPS** connector port
21. Connect the **external inputs cable** to the **Triggers** connector port on the DVR.
22. Connect the **system power cable** to the **Power** connector port on the DVR.
23. Insert the **15 amp fuse** into the system power cable fuse holder.
24. Power on the system, wait for the system to boot up, and then select the deployment mode (Deployment Mode on page 49).
25. Test the installation (Testing the System Installation on page 50).

2.1.1
Installation Overview UK



Reference Key
Supported Equipment
Optional Equipment

2.1.2
Installation Overview EU



2.2
Camera Installation

Motorola Solutions offers the following cameras with the M500 in-car video system:

- M5F front camera

  Figure 2: M5F Front Camera

  For more information, see Installing the M5F Front Camera.
- M5P camera

  Figure 3: M5P Camera

  For more information, see Installing the M5P Camera on page 27.
- M5R camera

  Figure 4: M5R Camera

  For more information, see Installing the M5R Camera on page 29.
- M5E camera

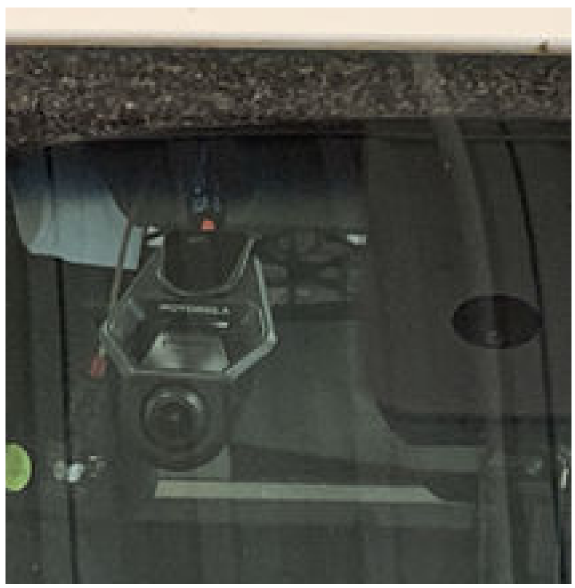
Figure 5: M5E Camera

For more information, see Installing the M5E Camera on page 30.

2.2.1
Installing the M5F Front Camera

The M5F is a small camera that mounts just behind the rearview mirror so it does not interfere with officer field of view.

Figure 6: M5F Camera Installed in a Vehicle



Procedure:

1. Clean the windshield surface using a microfiber cloth (or something similar), and alcohol or window cleaner.
2. Clean any residue from the puck, using alcohol or window cleaner.

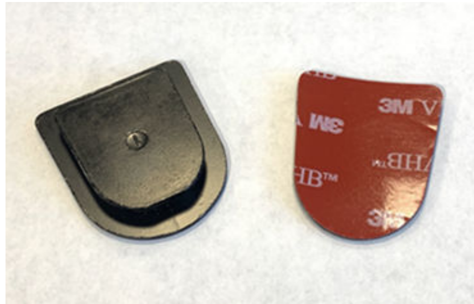
   IMPORTANT: DO NOT score the surface of the puck. Scoring the puck can keep it form properly adhering to the window.
3. Place the puck on the M5F front camera mount.

   TIP: Do not over-tighten the mount to the puck. Over-tightening causes a slight curve in the puck that keeps it from adhering properly to the windshield.
4. Determine where exactly on the windshield you want to mount the M5F.

   NOTE: Place the puck as high as possible on the windshield for a better viewing angle. See M5F Camera Placement on page 27 for more information.

5. If you are using the **3M™ VHB™ pre-cut tape** to install the puck:
   a. Peel the film off the clear side of the tape.

   Figure 7: 3M™ VHB™ pre-cut tape

   b. Place the clear side on the puck.
   c. Peel the film off the red side of the tape.
   d. Position the puck in the predetermined location on the clean windshield, and press the tape against the windshield.
6. **If you are using adhesive** to install the puck:
   a. (Optional) Heat the puck for 10 to 15 seconds, using a lighter or a hand-held butane torch, until it is warm to the touch.

   NOTE: Wipe off any black residue left from the lighter flame. The color of the puck changes slightly when it is heated.

   CAUTION: Handle any lighter or hand-held butane torch with care. Materials are hot and can cause harm.

   b. Carefully spray the primer on the puck AND the windshield (1 or 2 sprays to cover the area where you are mounting the puck).
   c. Let the primer dry.
   d. Apply a drop (5/16-inch or pea size) of adhesive to the puck.

   Figure 8: Puck with adhesive

   TIP: Apply enough adhesive that it spreads to every corner of the puck when you press the puck to the glass.

   e. Quickly press the puck with the camera attached to the windshield and hold it there for 60 seconds.

f. Leave the camera in place for at least 10 minutes to allow the adhesive to dry completely.

NOTE: If the puck does not successfully adhere to the windshield, replace the used puck with a new one and try again.

7. Connect the green HD-BNC cable to the camera then run it to the DVR location (but DO NOT connect the cable to the DVR).

   TIP: See Connecting HD-BNC Cables (Connecting HD-BNC Cables on page 17) for help connecting HD-BNC cables to the M500 system.

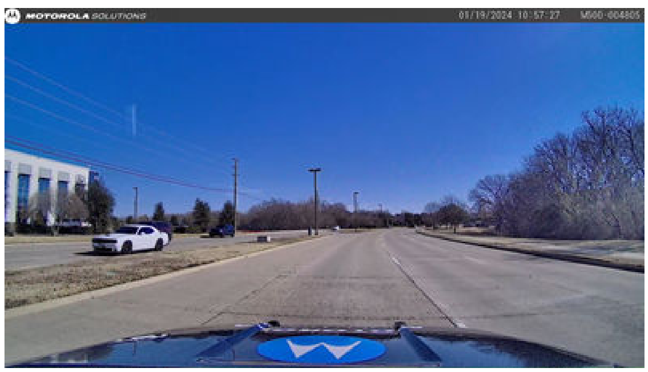
8. When instructed to do so in the full installation workflow (Installation Workflow on page 19), connect the green HD-BNC cable to the appropriate connector port on the DVR.

   IMPORTANT: Only connect cables when the full installation workflow (Installation Workflow on page 19) instructs you to. Connecting cables out of sequence can damage components.

2.2.1.1
M5F Camera Placement

The M5F camera placement is important for the system to capture Automatic License Plate Recognition (ALPR) information. Motorola Solutions suggests installing the camera higher on the windshield so that the hood of the vehicle is in the bottom 10% or less of the camera view.

Figure 9: M5F Camera View with Adjusted Placement



2.2.2
Installing the M5P Camera

The M5P is a compact, non-obtrusive camera with an integrated microphone that can be mounted on the cage or the roof liner.

Procedure:

1. Mount the M5P according to your installation plan.
2. Hold the M5P in place. Using a utility knife, cut the foam in the following places:

Figure 10: M5P Installed on the Cage

a. To the left of the camera, cut from the top to the bottom.
b. To the right of the camera, cut from the top to the bottom.
c. Beneath the camera, cut from one end to the other.

3. Holding the cut-out foam out of the way, install the M5P camera.
4. Fold and tuck the cut-out foam behind the camera and secure it between the headliner and the rubber padded roll bar.

Figure 11: M5P Secured to the Foam Bar on the Cage

5. Connect the purple HD-BNC cable to the camera and run it to the DVR location, but do **not** connect the cable to the DVR.
6. When instructed to do so in the Installation Workflow on page 19, connect the purple HD-BNC cable to the appropriate connector port on the DVR.

   For more information on connecting HD-BNC cables to the M500 system, see Connecting HD-BNC Cables on page 17.

2.2.3

Installing the M5R Camera

The M5R is a small camera that can be mounted on the rear windshield or the rear ceiling center. The chosen mounting position varies depending on the setup of the vehicle and the visibility needs of the driver.

Figure 12: M5R Installed

Procedure:

1. Mount the M5R according to your installation plan.

   The mounting locations vary depending on the type of vehicle:

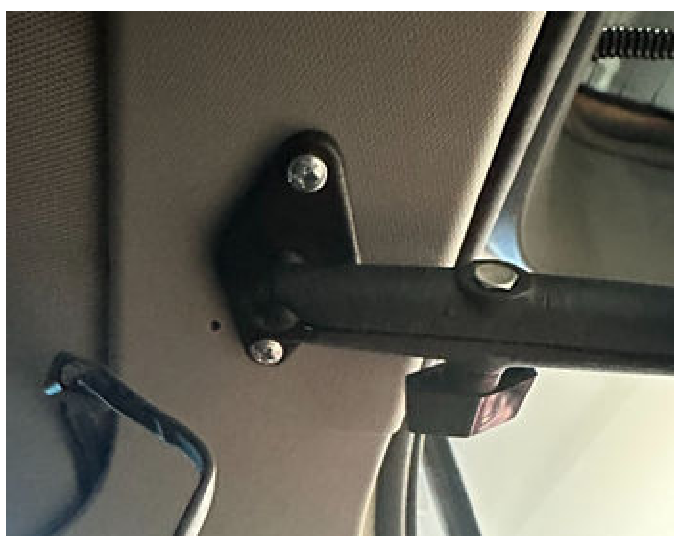
   - To mount the camera in an SUV, go to step 2.
   - To mount the camera in a sedan, go to step 3.

2. To mount the camera in an SUV, perform the following actions:

   a. Choose a secure location on the ceiling of the vehicle, inside the rear hatch door on the passenger side.

   b. In the chosen location, mount the RAM arm to the ceiling using two nuts and two bolts.

   Figure 13: M5R SUV Installation

   c. Attach the M5R camera with the pre-attached RAM ball to the RAM arm.

   d. Go to step 4.

3. To mount the camera in a sedan, perform the following actions:

a. Choose a location on the back window behind a headrest that minimizes obstructions in the field of view.

b. Clean the window surface using a microfiber cloth, or something similar, and alcohol or window cleaner.

c. In the chosen location, attach the diamond shaped RAM ball to the rear windshield with the corresponding 3M diamond shaped adhesive strip.

d. Attach the RAM arm socket.

e. Attach the M5R camera with the pre-attached RAM ball to the RAM arm socket.

4. Turn the camera slightly toward the driver side.

   NOTE: This action provides more visibility of the traffic behind the vehicle during a traffic stop.

5. Connect the appropriate HD-BNC cable to the camera and run it to the DVR location, but do **not** connect the cable to the DVR.

6. When instructed to do so in the Installation Workflow on page 19, connect the M5R HD-BNC cable to the appropriate connector port on the DVR.

   For more information on connecting HD-BNC cables to the M500 system, see Connecting HD-BNC Cables on page 17.

2.2.4
Installing the M5E Camera

The M5E is a 1080p HD high resolution camera that is weatherized and can be mounted on the exterior of a vehicle.

The M5E is an exterior camera facing away from the vehicle in one of the following locations:

- Left exterior side of the vehicle
- Right exterior side of the vehicle

Figure 14: M5E Installed on the Right Exterior Side of the Vehicle

NOTE: Depending on your installation plan and DVR location, you may need to drill new holes or reuse existing holes to run the HD-BNC cable to the DVR location.

Procedure:

1. Mount the M5E according to your installation plan.
2. Attach the appropriate bracket to the light bar.
3. Install the side rail mount to the bracket.

   NOTE: The side rail mount can be installed without a light bar.

4. Connect the appropriate HD-BNC cable to the camera.
5. Run the HD-BNC cable from the camera, through the roof rack channel on the top of the vehicle, and to the DVR location, but do **not** connect the cable to the DVR.

   Figure 15: Roof Channel Rack

6. When instructed to do so in the Installation Workflow on page 19, connect the M5E HD-BNC cable to the appropriate connector port on the DVR.

   For more information on connecting HD-BNC cables to the M500 system, see Connecting HD-BNC Cables on page 17.

2.3
Installing the M500 DVR Display

Typically, you mount the M500 DVR Display on a RAM® ball mount.

TIP: Where you mount the M500 DVR Display varies from vehicle to vehicle. For more information, see the mounting instructions included with the brackets.

Figure 16: M500 DVR Display Installed

Procedure:

1. Install one of the RAM® balls from the mount on the back of the display.
2. Connect the appropriate HD-BNC cable to the back of the display.

   NOTE: See Connecting HD-BNC Cables (Connecting HD-BNC Cables on page 17) for help connecting HD-BNC cables to the M500 system.

3. Install the display bracket on the passenger side visor.
4. Mount the other RAM® ball to the bracket.
5. Connect the mount double socket to the RAM® ball on the bracket.
6. Connect the RAM ® ball mount on the back of the display to the double socket.
7. Run the display HD-BNC cable to the DVR location, but DO NOT connect it to the DVR.
8. When instructed to do so in the full installation workflow (Installation Workflow on page 19), connect the display HD-BNC cable to the DVR.

   IMPORTANT: Only connect cables when the full installation workflow (Installation Workflow on page 19) instructs you to. Connecting cables out of sequence can damage components.

2.4
M500 DVR Installation

The M500 In-Car Video System supports up to five cameras:

- Front (M5F)

- Cabin (M5P)
- Right (M5E)
- Rear (M5R)
- Left (M5E)

NOTE: The camera ports for the M5E cameras are marked as **LS** and **RS**. See Installation Overview UK on page 22 or Installation Overview EU on page 23 for more information.

The M5P, M5R, and M5E cameras can connect to any camera port except for the M5F port.

NOTE: The 5-camera system also has a USB port on the back of the DVR to support an RFID reader. See RFID Installation on page 45 for more information.

Where you mount the DVR depends on agency preference as well as the vehicle where you are installing the system. The most common locations for mounting the DVR include:

- In the console
- In the trunk or the back of the vehicle

2.4.1
Installation Examples

The DVR can be installed in the console:

Figure 17: DVR Installed in a Console

Table 1: DVR Installation in a Console

Item	Description
1	Console
2	DVR

The DVR can be installed in the back or trunk of the vehicle:

Figure 18: DVR and LTE Router in a Trunk

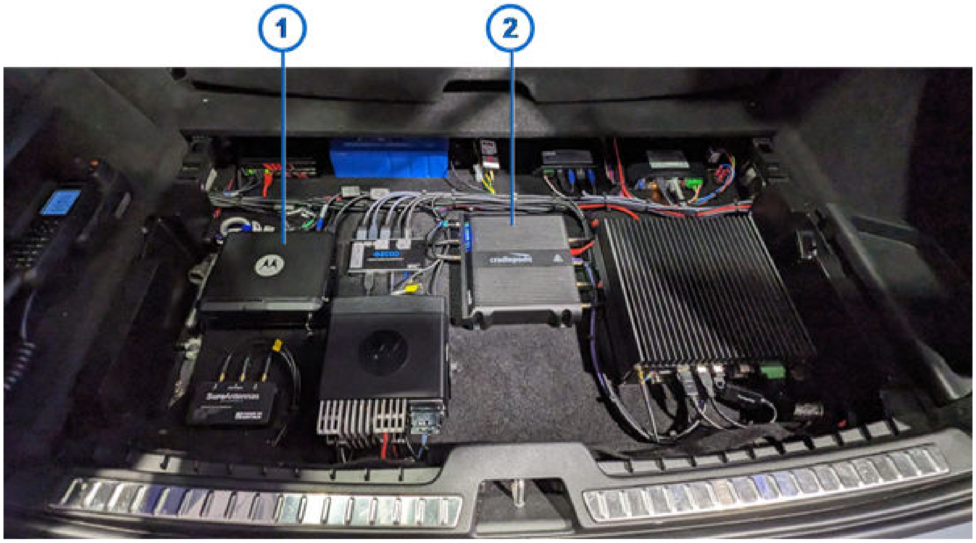


Table 2: DVR and LTE Router Installation in a Trunk

Item	Description
1	M500 DVR
2	LTE router

Figure 19: DVR and MikroTik Groove in a Trunk

Table 3: DVR and MikroTik Groove Installation in a Trunk

Item	Description
1	MikroTik Groove radio
2	M500 DVR

2.4.2
Installing the DVR

Procedure:

1. Install the DVR bracket.

   NOTE: For information about installing the DVR bracket for a particular vehicle, see the mounting instructions included with the bracket.

2. Position and loosely install the DVR, but do **not** connect cables to it.

   IMPORTANT: Only connect cables when the Installation Workflow on page 19 instructs you to. Connecting cables out of sequence can damage components.

3. When instructed to do so in the Installation Workflow on page 19, connect cables to the back of the DVR.

2.5
Wireless Radio Installation

IMPORTANT: Motorola Solutions recommends that wireless radios be configured before you mount them in the vehicle. Typically, radios purchased from Motorola Solutions are configured before they leave the factory. If you have chosen to configure your own wireless radios, before starting the configuration process, contact Motorola Solutions Customer Service for instructions.

Motorola Solutions provides the MikroTik Groove as the system wireless radio (MikroTik Groove on page 38):

Figure 20: MikroTik Groove

You should mount the wireless radio where its LEDs can be seen.

NOTE: No mounting hardware is included with the wireless radio.

2.5.1
Installing a Wireless Radio

Procedure:

1. Mount the wireless radio according to your installation plan, but DO NOT connect any cables.

   IMPORTANT: Make sure you mount the wireless radio in a position where you can easily see its LEDs. The LEDs are needed for diagnostic purposes.
2. Mount the wireless radio antenna (Mounting the Wireless Radio Antenna on page 36).
3. Run the antenna cable to the wireless radio then connect it.

   WARNING: DO NOT plug in the wireless radio PoE connector (Ethernet connection) and power up the radio WITHOUT an antenna connected. Powering up the radio without an antenna can damage the radio.
4. When instructed to do so in the full installation workflow, connect an Ethernet cable from the wireless radio to the red connector port on the DVR.

   See Installation Workflow on page 19 for more information.

2.5.2
Mounting the Wireless Radio Antenna

Typically, you mount the radio antenna on the roof or on the trunk of the vehicle.

IMPORTANT: Motorola Solutions recommends that you keep about 9 inches between the wireless radio antenna and other antennas. You must keep a minimum of 6 inches between antennas.

You route the antenna cable into the vehicle to the location where you plan to mount the wireless radio. You can choose from two different types of radio antenna mounts:

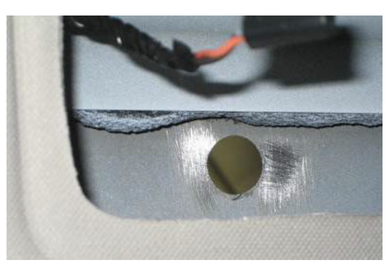
- Through-hole antenna (NMO mount) which requires that you drill a 3/4-inch hole through the vehicle roof
  The example installation in this section shows an NMO mount antenna
- Magnetic mount which does not require that you drill a hole

  IMPORTANT: Motorola Solutions recommends that you use the NMO mount antenna because it provides better signal strength.

Procedure:

1. Access the roof of the vehicle in your preplanned location (for example, through the rear seat dome light or by removing the headliner).
2. Drill a 3/4-inch hole for the antenna cable.
3. Sand the paint off the metal around the interior hole to ensure a good ground.

   Figure 21: Interior Hole

   
4. Remove the NMO nut from the radio antenna cable.

5. Starting with the radio-connector end, feed the antenna cable through the exterior hole in the vehicle until only the antenna mount (NMO-connector) end of the cable remains outside the vehicle.

   Figure 22: Radio Connector End

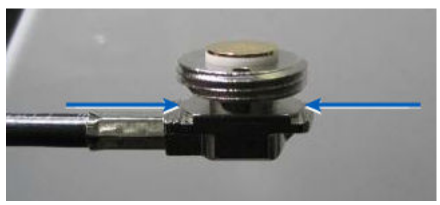
6. Tilt the antenna mount end to guide it through the exterior hole, then center the flange in the hole.

   Figure 23: Antenna Mount End

7. Make sure that the antenna mount ridges are in contact with the bare metal (from your sanding) inside the vehicle.

   Figure 24: Antenna Mount Ridges

8. On the antenna mount outside the vehicle, reattach the NMO nut, with the O-ring facing down to create a seal.

   Figure 25: NMO Nut and O-Ring

9. Tighten the nut with a 24mm or 15/16-inch open-end wrench, making sure that the mount stays centered in the hole and does not turn and kink the cable.

   TIP: To keep the cable from turning and kinking, use needle-nose pliers in the holes on the antenna mount to anchor the mount while you tighten the NMO nut.

Figure 26: Tightening the NMO Nut

⚠ **WARNING:** DO NOT tighten the NMO nut using channel-lock pliers. Tightening the NMO nut using channel-lock pliers can damage the nut and cause an unreliable wireless signal.

10. Inside the vehicle, route the radio-connector end of the antenna cable to the location where mounted the radio, making sure that you leave enough slack to avoid kinks.
11. Attach the antenna connector to the radio.

Figure 27: Antenna Connector and Radio

⚠ **WARNING:** DO NOT plug in the wireless radio's PoE connector (Ethernet connection) and power up the radio WITHOUT an antenna connected. Powering up the radio without an antenna can damage the radio.

2.5.3 MikroTik Groove

⚠ **WARNING:** Do not connect the wireless radio Ethernet connection or power up the radio without an antenna connected. Powering up the radio without an antenna can damage the radio.

Figure 28: MikroTik Groove

To verify that the MikroTik Groove wireless radio is working properly, check the LEDs on the side of the device:

Figure 29: MikroTik Groove LEDs

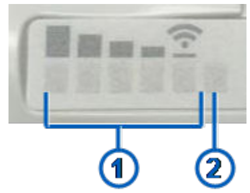


Table 4: MikroTik Groove Radio Information

Item	Description
1	Signal strength
2	Power/Connectivity

- **Signal strength LEDs**: When lit, shows that the radio is associated with the agency's access point
  - All of the LEDs on the Groove radio are green when lit; The signal between the radio and the access point is strongest when all five signal-strength LEDs are lit
  - The radio beeps once to indicate it is receiving power and a second time to indicate that it is booted up
- **Power/Connectivity LED**: When lit, shows that the radio has power and is connected to the M500 system

  NOTE: To determine whether the Groove radio is communicating successfully with the M500, after the M500 is configured, press **Home** on the Display Control Panel. On the **Home** menu, touch **Status**, then **Network**, then scroll to **My Vehicle Radio**. Verify that the **Access Connected** value is **Yes**.

  NOTE: The MikroTik PoE Adapter provides additional power to the MikroTik Groove. See Installing the Wireless Radio PoE Adapter on page 39 for more information.

2.5.3.1
Installing the Wireless Radio PoE Adapter

The wireless radio PoE adapter is a single adapter that provides power to the wireless radio.

When and where to use:

IMPORTANT: You only need to install the adapter if you are installing the M500 system with a wireless radio.

The adapter requires two cables:

- An Ethernet cable that connects the DVR to the DVR Ethernet port on the adapter
- An Ethernet cable that connects the wireless radio to the other Ethernet port on the adapter

Figure 30: MikroTik PoE Adapter

Procedure:

1. Determine the best place in the vehicle to install the adapter.
2. Connect Ethernet cable 1 to the adapter port with the LEDs.
3. Connect the other end of Ethernet cable 1 to the wireless radio.
4. Connect Ethernet cable 2 to the adapter port marked **DVR**.
5. When instructed to do so in the full installation workflow (Installation Workflow on page 19), connect the other end of the Ethernet cable to the red switch port on the DVR.

   To verify that the adapter is applying power, check the LEDs on the side of the device. The LEDs turn on but will not blink.

Figure 31: MikroTik Adapter LEDs

2.6
Microphone Installation

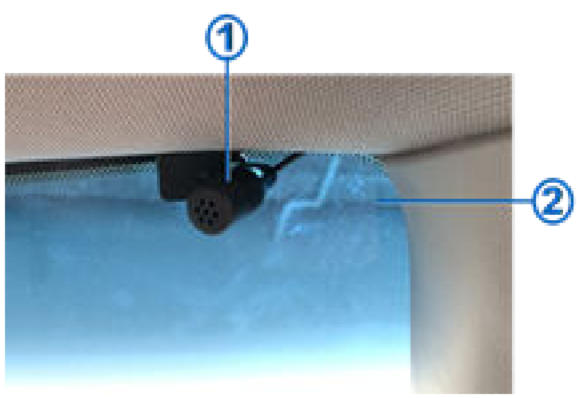
Motorola Solutions provides two types of microphones for the M500 in-car video system:

- Stand-alone cabin (Installing the Stand-Alone Cabin Microphones on page 41)
- Integrated (M5P camera) (M5P Camera Integrated Microphone on page 42)

2.6.1
Installing the Stand-Alone Cabin Microphones

The stand-alone cabin microphones are omnidirectional, wired, in-car microphones. They should be mounted on either top corner of the windshield for best results.

Figure 32: Stand-Alone Cabin Microphone Installed



Item	Description
1	Cabin microphone, installed
2	Vehicle windshield, top passenger corner

Procedure:

1. Mount the first cabin microphone at the top corner of the driver side windshield.
2. Mount the second cabin microphone at the top corner of the passenger side windshield.
3. Run the cabin microphone cables to the DVR location, but do **not** connect the cables.
4. When instructed to do so in the Installation Workflow on page 19, connect the cabin microphone cables to the appropriate connector ports on the DVR.

   WARNING: Ensure that you connect the cabin microphone cables to the DVR before you power the DVR ON. Connecting the cabin microphone cable after you power ON the DVR can damage the cabin microphone connector ports on the DVR.

2.6.2
M5P Camera Integrated Microphone

The M5P camera has an integrated, omnidirectional microphone. It is a compact camera/microphone that can be mounted on the cage or the roof liner.

Figure 33: M5P Cabin Camera Integrated Microphone

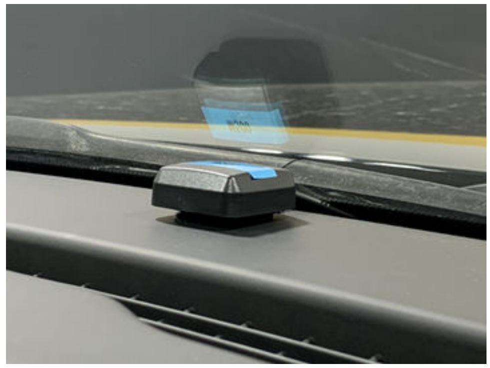
NOTE: There is no further installation required for the integrated microphone once the M5P camera is installed according to the Installation Workflow on page 19.

2.7
Installing the GPS Antenna

You should mount the GPS antenna with line-of-sight to the sky, for example, exterior roof, exterior trunk, or interior dashboard.

NOTE: Motorola Solutions recommends that you mount the GPS antenna on top of the dashboard as close to the center as possible, at least 3 inches away from any other objects. The GPS antenna should not be blocked by any window tint, if possible.

Figure 34: GPS Antenna



Procedure:

1. Mount the GPS antenna according to your installation plan.
2. Run the GPS antenna cable to the DVR location, but do **not** connect the cable.
3. When instructed to do so in the Installation Workflow on page 19, connect the GPS antenna cable onto the GPS connector port on the DVR.

   NOTE:
   Due to different vehicle and windshield manufacturers, in some cases when the GPS antenna is mounted on the dashboard, the antenna does not consistently lock in on satellites. This can cause the M500E to record incorrect speeds and possibly trigger an erroneous recorded event with the over-speed trigger.

   In this case, remount the GPS antenna outside the vehicle on the roof or the trunk in line-of-sight to the sky.

2.8
LTE Router Installation

The LTE router uploads recorded events from the vehicle to the agency evidence management system over a cellular connection.

If present in the system, the LTE router connects to the DVR.

Figure 35: LTE Router

Motorola Solutions supports the following Sierra Wireless® routers for the :

- Sierra Wireless® AirLink® RV 55 Industrial LTE-A Pro Router

- Sierra Wireless® AirLink® MP70 LTE Router
- Sierra Wireless® AirLink® XR60 Router
- Sierra Wireless® AirLink® XR80 Router

NOTE: Motorola Solutions will support other routers in future versions of the M500 in-car video system.

2.8.1 Installing the LTE Router

Procedure:

1. Mount the LTE router according to your installation plan.
2. Locate one Ethernet cable included in the DVR installation kit.
3. When instructed to do so in the Installation Workflow on page 19, connect one end of the Ethernet cable to the appropriate Ethernet port on the router.
4. When instructed to do so in the Installation Workflow on page 19, connect the other end of the Ethernet cable to one of the switch ports on the DVR.

2.9 MDC Installation

If present in the system, the MDC (mobile data computer) or other laptop computer connects to the M500 DVR.

The computer only connects into the system through the switch ports on the M500.

2.9.1 Installing the MDC

Procedure:

1. Identify the appropriate location in your vehicle for the MDC installation and install the MDC in the vehicle.
2. Locate one Ethernet cable included in the DVR installation kit.
3. Connect one end of the Ethernet cable to the appropriate Ethernet port on the MDC.
4. When instructed to do so in the Installation Workflow on page 19, connect the other end of the Ethernet cable to one of the switch ports on the DVR.

2.10
RFID Installation

The Radio-Frequency Identification (RFID) reader connects to the back of the M500 DVR.

Only connect the RFID reader to the USB port on the back of the DVR.

IMPORTANT: Do not connect the RFID reader into the USB port on the front of the DVR.

When instructed to do so in the installation workflow, connect the RFID reader to the USB port on the back of the DVR.

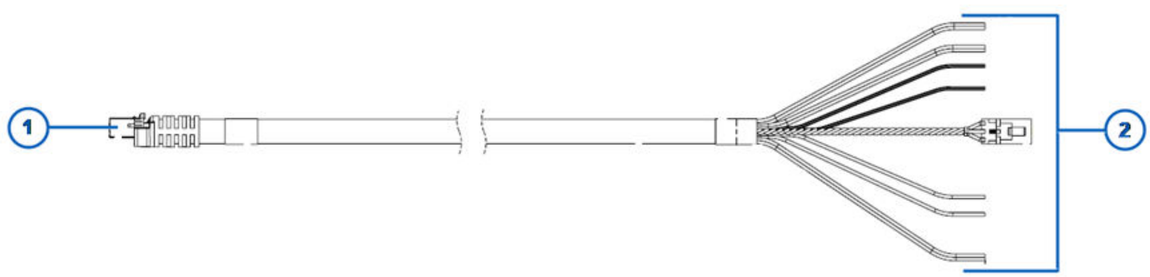
NOTE: See Installation Workflow on page 19 for more information.

2.11
Cable Connections for External Inputs

The external inputs harness connects the M500 to the external device inputs, for example, ignition, emergency lights, and siren.

WARNING: Only connect cables when the Installation Workflow on page 19 instructs you to. Connecting cables out of sequence can damage components.

Figure 36: External Inputs Harness



Item	Description
1	DVR connector
2	External input sense wires

The sense wires in the external inputs cable monitor the external device inputs for voltage (8–12 V) and no voltage (0 V). When an external input wire senses voltage from an external device, the DVR starts a recorded event, if configured to do so in the agency evidence management system. 0 V signals to the M500 that it should prompt the user to stop the event, if configured to do so.

The sense wire for the ignition signals the DVR to start the shutdown sequence and power off or it signals the DVR to power up.

TIP: Where possible, use butt splices when you connect the wires to ensure good connection and consistent voltage.

Connect each sense wire in the external inputs cable to the corresponding external device input:

- **Ignition (white)**: The white wire senses whether the ignition is on or off. The presence of voltage on the sense wire also signals to the system to add the external device activation information to the metadata file.

  Connect the white wire to a feed (switched ignition accessory wire) that provides positive 12 V when the vehicle ignition is ON and 0 V when the vehicle ignition is OFF.
- **Emergency lights (blue/white)**: The blue wire with the white stripe senses whether the emergency lights have been activated

  Connect the blue wire with the white stripe to a positive 12-volt switched power source that provides voltage when the overhead lights are activated.
- **Siren speaker (-) (light green)** and **Siren speaker (+) (pink)**: The pink and light green wires sense whether the siren has been activated

  Connect the light green wire to the siren speaker ground (-) feed and the pink wire to the positive (+) siren speaker feed. When the siren is activated, the siren input senses the voltage using the siren speaker.
- **Auxiliary 1 (blue)** and **Auxiliary 2 (red)**: Each of the wires senses whether an auxiliary device has been activated

  Connect the wire to a source that provides positive 12-volt switched power when an auxiliary device is activated. For example, when the electric release button is pressed on a shotgun mount or an electric K-9 kennel door is released.
- **Brakes (black/white)**: The black wire with the white stripe senses whether the brakes are being applied

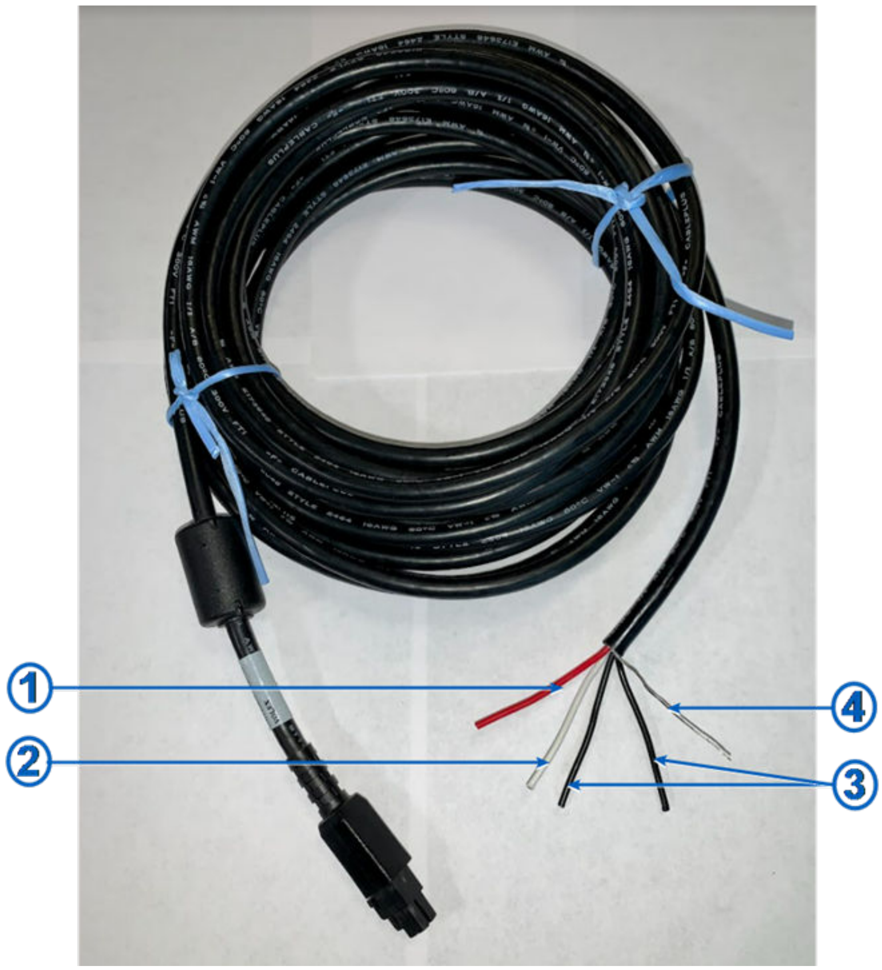
  Connect the black wire with the white stripe to a positive 12-volt switched power source that is activated when the brake pedal is applied. When the brakes are applied, the wire should get 1–12 V.

TIP: Each of the sense wires in the external inputs cable is labeled with the name of the external device input it should connect to. If you need to cut the cable, for example, because it is too long for the space you have, bind the cut labeled ends together and save them to serve as the key for the connections.

2.12
Connecting the System Power Cable to the Vehicle Battery

The system power cable connects the M500 to the vehicle battery.

Figure 37: System Power Cable



Item	Description
1	Power wire (red)
2	Battery backup/iUPS power cable (white) For future use
3	Ground wires (black)
4	Drain wire (silver)

WARNING: Only connect cables when the Installation Workflow on page 19 instructs you to. Connecting cables out of sequence can damage components.

Procedure:

1. Determine how you will connect the power cable to vehicle battery power.

The options are as follows:

- Leads connected from the battery posts to a predetermined location in the vehicle, for example, a fuse block; power cable also connected to the fuse block
- Power cable connected directly to the vehicle battery

2. Prepare to connect the power cable to battery power.

   The options are as follows:

   - Connect the leads to the battery posts and run them to the predetermined location, for example, the fuse block.
   - Determine how to run the power cable into the engine compartment and to the battery.

   IMPORTANT: Use the firewall holes and grommets provided by the vehicle manufacturer to route the cable to the battery. Consult your vehicle manufacturer's documentation for specific information.

3. Run the DVR connector end of the power cable to the DVR location, but do **not** connect the cable.
4. Run the power wires end of the power cable to one of the following:

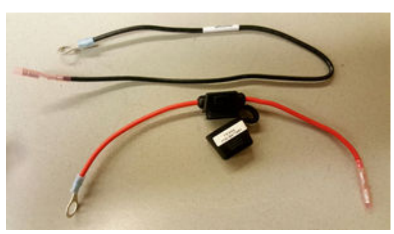
   - The location of the leads, for example, the fuse block
   - The vehicle battery location

   IMPORTANT: Leave an extra 1 to 2 feet of slack on the power cable for a drip loop.

5. Wind up and tie off or cut the excess power cable.
6. Strip back the sheath on the power cable, then on the red (positive) wire and one of the black (ground) wires.

   IMPORTANT: Do **not** strip the white wire or other black wire. These wires are for future use with an auxiliary battery backup.

7. Tape or heat-shrink the ends of the white wire and the other black wire, and then tape them to the power cable ensuring that they are not exposed.
8. Twist the stripped black (ground) wire and silver drain wire together, then crimp them to the provided black (ground) wire extension with the attached ring terminal.

9. Crimp the red (positive) wire to the provided red (positive) 15 amp fuse holder with the attached ring terminal.
10. Connect the red (positive) and black (ground) ring terminals to one of the following:

    - The leads from the battery, for example, at the fuse block
    - The vehicle battery posts

11. When instructed to do so in the Installation Workflow on page 19), connect the DVR connector end of the power cable to the Power connector port on the DVR.

12. When instructed to do so in the Installation Workflow on page 19, insert the fuse in the fuse holder.

    WARNING: Do **not** insert the fuse until you have finished installing and connecting all the system equipment. Inserting the fuse before you have finished the full installation can damage the system.

2.13 Deployment Mode

The M500 system is configured, by default, for the integrated Smart Power Switch (SPS) and other components. However, after the initial boot up of the M500, the installer chooses whether or not the system should be configured using the following components:

- Display
- M5F
- Integrated SPS

The **Display**, **M5F**, and **Integrated SPS** check boxes are checked by default. To proceed without one or more of these components, clear the appropriate check boxes.

NOTE: If the installer clears the **Integrated SPS** check box, the system initializes without an SPS configuration.

2.13.1 Selecting the Deployment Mode

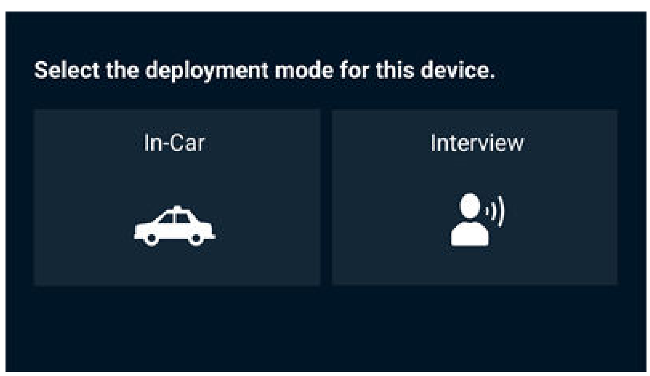
When the installer powers on the M500 for the first time, they select the deployment mode for the system being installed.

Procedure:

1. To power on the M500, press and hold the **On** button for 2 seconds.

   After the M500 has powered on, the initial deployment mode screen appears.

   Figure 38: Deployment Mode Screen

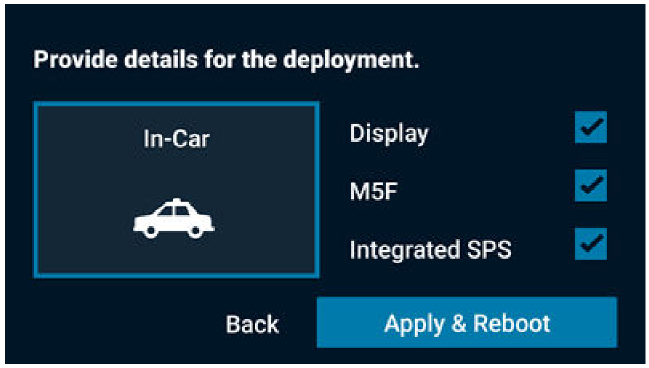
   

   NOTE: **Interview** deployment mode is optimized for a fixed room environment and not recommended for an in-vehicle installation.

2. Select the **In-Car** deployment mode.
3. Touch **Next**.

   The **Provide details for the deployment** screen appears.

Figure 39: In-Car Deployment Mode Details



NOTE: The **Display**, **M5F**, and **Integrated SPS** check boxes are checked by default. To proceed without one or more of these components, clear the appropriate check boxes.

4. Check or clear the appropriate check boxes for the installation.

   IMPORTANT: If you clear the **Integrated SPS** check box, the system initializes without an SPS configuration.

5. Touch **Apply & Reboot**.

   A confirmation message appears and the M500 reboots.

2.14 Testing the System Installation

IMPORTANT: Make sure that you have finished all the steps in the recommended installation workflow (Installation Workflow on page 19) before you test the installation.

Procedure:

1. Verify that there is power to the system.

   The Display Screen should show Live View.

2. Test the touch screen.
   a. Press **Home** on the M500 DVR Display Control Panel.
   b. Touch any icon on the Display Screen.
   c. Verify that the correct screen opens.
3. Verify that the M500 DVR Display Screen shows views from all of the installed cameras.
   a. Touch the Live View screen to see the on-screen arrows.
   b. Touch an arrow to move to another camera view.
4. Verify that the external inputs are correctly connected:
   a. Press **Home** on the Display Control Panel, then touch **Status** > **System** on the Display Screen.
   b. Activate each input you connected, for example, **Siren**.
   c. Verify that each input shows **On** when you activate it.
5. Verify the wireless radio connection, as applicable:
   a. Verify the LEDs on the wireless radio for power and connectivity.

      See *MikroTik Groove* (MikroTik Groove on page 38) for information on the MikroTik Groove radio LEDs.

b. Press **Home** on the Display Control Panel, then touch **Status** > **Network** on the Display Screen and scroll to **My Vehicle Radio**.
c. Verify that the **Access Connected** value is **Yes**.
d. Verify that the Wi-Fi icon appears on the Display Screen.

6. Create one or more test recorded events:
   a. Press **Record** on the Display Control Panel.
   b. Allow the event to continue for at least 15 seconds.
   c. Press **Stop** on the Display Control Panel to stop the event.
7. Test recorded event playback with audio:
   a. Press **Home** on the Display Control Panel, then touch **Review** on the Display Screen.
   b. Touch an event to select it from the list, then use the player controls to review.
8. If you have a configuration from the agency evidence management system, load it on the M500 DVR and verify the configuration items.

Appendix A

Standard Installation Configurations

This appendix includes information that will help you to install the Motorola Solutions M500 in-car video system, body-worn cameras, and related components. Any other configurations may require custom mounting hardware and installation services.

Please contact your Motorola Solutions representative for any information that is not addressed in this document.

DVR Locations

Where you mount the DVR depends on agency preference as well as the vehicle where you are installing the system. The most common locations for mounting the DVR include in the trunk or the console.

Table 5: DVR Brackets

Bracket Type	Part Number
DVR Mounting Bracket (Console)	WGP02919-KIT
DVR Mounting Bracket (Trunk)	WGP02920-KIT

Camera Locations

Where you mount the cameras depends on agency preference as well as the vehicle where you are installing the system. The most common locations for mounting the cameras are as follows:

Table 6: Camera Brackets

Camera Type	Recommended Mount Location
M5F	Windshield (mount included)
M5P	Cage or visor post (brackets included)
M5R	Rear windshield (Mounting hardware included)
M5E	Top of car door (brackets included)

NOTE: The M5P camera can be attached to the passenger seat visor bracket in 2025 or later Ford Explorer models using bracket WGP02225-550-KIT.

DVR Display Locations

The intended attachment location for the DVR Display is the passenger visor post. The installer uses the bracket designated for the vehicle make and model as specified in M500 Bracket and Vehicle Information on page 54.

Supported Vehicles

Mounting brackets are maintained for the current most popular vehicles used in public safety:

- Chevrolet Tahoe (+GMC Yukon) (2005-2025)
- Chevrolet Silverado (2005-2025)
- Dodge Durango (2010-2025)

- Ford Police Interceptor Utility (Explorer) (2010-2025)
- Ford F150 Police Responder (2005-2025)

In addition, brackets are provided for the following older or discontinued vehicles:

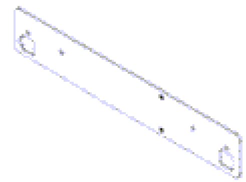
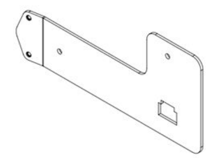
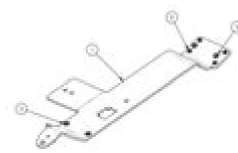
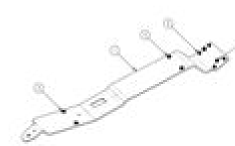
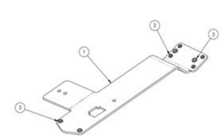
- Chevrolet Caprice (2011-2017)
- Chevrolet Impala (2014-2020)
- Dodge Charger (2006-2023)
- Dodge RAM (2019-2021)
- Ford Taurus (2010-2019)
- Ford Expedition (2006-2021)
- Ford Crown Victoria (2005-2012)

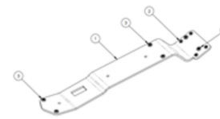
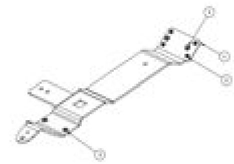
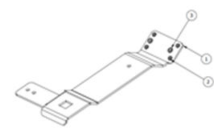
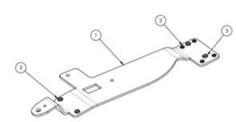
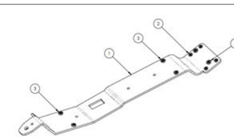
NOTE: Any vehicle make, model, or year not listed must be assessed by Motorola Solutions Sales Engineering or by your agency's installation team to determine an appropriate installation method.

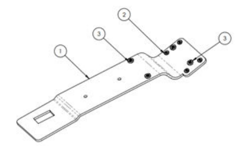
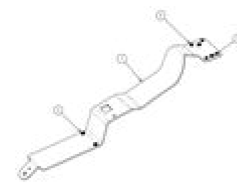
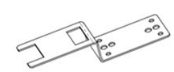
Appendix B

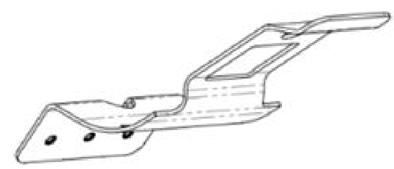
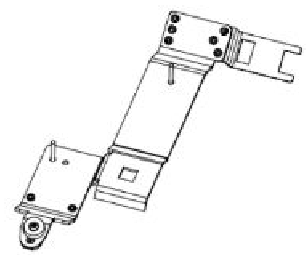
M500 Bracket and Vehicle Information

Table 7: M500 Brackets and Part Numbers

Part Number	Description	KIT	KIT 2	Image
WGP01459-003	BRKTKit4REDisplayFord-CrownVic2005(b)+	• WGP01459-003 • WGP01459-HDW	N/A	
WGP01459-018	Brkt 4RE/M500 Display Generic Fit	N/A	• WGP01459-018 • WGP02225-SCREW-KIT	
WGP02225-100	Brkt4RE Disp/HMic/CamVr 07-14Tah11Expl	• WGP02225-100 • WGP02225-201 • WGP02225-SCREW-KIT • WGP515-002	• WGP-02225-100 • WGP02225-SCREW-KIT • WGP515-002	
WGP02225-102	BRAKCET, 4RE DISPLAY, 2015+ TAHOE	• WGP02225-102 • WGP02225-201 • WGP02225-SCREW-KIT • WGP515-002 • WGD00123-102	• WGP02225-102 • WGP02225-SCREW-KIT • WGP515-002	
WGP02225-121	Brkt4RE DispOnly 07-14 TAHOE 2011+ EXPL	N/A	• WGP02225-121 • WGP02225-SCREW-KIT	

Part Number	Description	KIT	KIT 2	Image
WGP02225-122	Brkt4RE DispOnly Light-Modified 15 TAHOE	N/A	• WGP02225-122 • WGP02225-SCREW-KIT	
WGP02225-130	BRKT 4RE Disp/HiFi-Mic/CamVisor +20 Explo	• WGP02225-130 • WGP02225-201 • WGP02225-SCREW-KIT • WGP515-002	• WGP02225-130 • WGP02225-SCREW-KIT • WGP515-002	
WGP02225-131	Brkt4RE DispOnly Light-Modified 20 Expl	• WGP02225-131 • WGP02225-201 • WGP01552-002-Z • WGP01780-005	• WGP02225-131 • WGP02225-SCREW-KIT	
WGP02225-200	Brkt4RE Disp/HMic/ZCam +Charger	• WGP02225-220 • WGP02225-201 • WGP02225-SCREW-KIT • WGP515-002	• WGP02225-220 • WGP02225-SCREW-KIT • WGP515-002	
WGP02225-230	BRACKET,IN-CAR,2021+TAHOE	• WGP02225-230 • WGP02225-201 • WGP02225-SCREW-KIT • WDG00123-102	• WGP02225-230 • WGP02225-SCREW-KIT	

Part Number	Description	KIT	KIT 2	Image
WGP02225-231	BRACKET,IN-CAR,2021+TA-HOE,LIGHTBAR	• WGP02225-231 • WGP02225-201 • WGP02225-SCREW-KIT • WGD00123-102	• WGP02225-231 • WGP02225-SCREW-KIT	
WGP02225-600	Bracket, 4RE, Display, 2015 F150	• WGP02225-600 • WGP02225-201 • WGP02225-SCREW-KIT • WGP515-002	• WGP02225-600 • WGP02225-SCREW-KIT • WGP515-002	
WGP02225-201	Hi-Fi Microphone bracket	N/A	N/A	
WGP02225-SCREW-KIT	• WGP01552-002-Z: ScrewPPHSelfDrilling#8x1.25"BlackZinc • WGP01773-Z: RubberBumper#8Screw.94Dx.38H Ketstone#723 • WGP01780-005: Screw, #6-32 x 5/16", PPH, Zinc, Black • WGP316: FlatHeadScrew#8-32x1/4"LUndercut BK Zinc • WGP385-807: Screw, PPH, #8-32 x .437L, Black Zinc • WGP515-002: MIRROR BUTTON, BRACKET MOUNT (non-GLASS)	N/A	N/A	N/A

Part Number	Description	KIT	KIT 2	Image
WGP01459-HDW	• WGP01545-002: Screw, PPH, #8-32 x 1.5", Clear Zinc • WGP01546-001-Z: ScrewHWHSelfDrilling#10x.750"ClearZinc • WGP01552-002-Z: ScrewPPHSelfDrilling#8x1.25"BlackZinc	N/A	N/A	N/A
WGP02225-550	Bracket, Camera Visor Post, 2025+ Ford	• WGP02225-507 • WGP02225-550 • WGP01780-001 • WGP385-807	N/A	
WGP02225-650	Bracket, Display + HiFi-Mic + Cam, Visor, 2025+ Explorer	• WGP02225-650 • WGP02225-201 • WG02225-SCREW-KIT • WGP515-002	• WGP02225-650 • WGP02225-SCREW-KIT • WGP515-002	

Display Brackets for Particular Vehicles

M500 Display Brackets are designed to attach to the passenger visor post of a vehicle. Each display bracket includes mounting points for a WiFi Base (WGA00635-KIT) and HiFi microphone using the bracket (WGP02225-201) that is included with the HiFi microphone.

If the vehicle has an internal light bar installed, an alternate bracket is available which accounts for this obstruction. Any bracket that has an asterisk(*) may require modification for a successful installation. If you need further assistance, contact your Motorola Solutions representative.

Table 8: Display Brackets by Vehicle

Make	Model	Year	Bracket Part Number	If light bar is installed
Chevy/GMC	Silverado	2021-2023+	WGP02225-230-KIT2	WGP02225-231-KIT2
		2015-2020	WGP02225-102-KIT2	WGP02225-122-KIT2

Make	Model	Year	Bracket Part Number	If light bar is installed
		2005-2014	WGP02225-100-KIT2	WGP02225-121-KIT2
	Tahoe/Yukon	2021-2023+	WGP02225-230-KIT2	WGP02225-231-KIT2
		2015-2020	WGP02225-102-KIT2	WGP02225-122-KIT2
		2005-2014	WGP02225-100-KIT2	WGP02225-121-KIT2
	Caprice	2011-2017	WGP02225-100-KIT2	WGP02225-100-KIT2*
	Impala	2014-2020	WGP02225-100-KIT2	WGP02225-121-KIT
Dodge	Charger	2010-2023	WGP02225-200-KIT2	WGP02225-200-KIT2*
		2006-2009	WGP01459-018-KIT2	WGP01459-018-KIT2*
	Durango	2010-2023+	WGP02225-102-KIT2*	WGP02225-122-KIT2*
	RAM	2019-2021	WGP02225-200-KIT2	WGP02225-200-KIT2*
Ford	Taurus	2010-2019	WGP02225-100-KIT2	WGP02225-121-KIT2*
	Expedition	2015-2021	WGP02225-600-KIT2	WGP02225-600-KIT2*
		2006-2014	WGP02225-100-KIT2	WGP02225-121-KIT2
	Explorer	2025+	WGP02225-650-KIT2	not yet available
		2020-2024	WGP02225-130-KIT2	WGP02225-131-KIT2
		2010-2019	WGP02225-100-KIT2	WGP02225-121-KIT2
	F150	2015-2023+	WGP02225-600-KIT2	WGP02225-600-KIT2*
		2005-2014	WGP02225-100-KIT2	WGP02225-121-KIT2
	Crown Victoria	2005-2012	WGP01459-003-KIT	WGP01459-003-KIT